

CV

Dr. Jaspreet Hira

16+ Years Experience

Current Designation: **Professor**

Department of Mechanical Engineering

Maharishi Markandeshwar (Deemed to be University) NAAC A⁺⁺

Mullana, Ambala

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Mobile Number: +91-9560149127

H Index **5**, I-10 index **3** and Total Citation: **104**

Scopus Author: 57189442197

Google Research: <https://scholar.google.com/citations?user=LsoFj4AAAAAJ&hl=en>

LinkedIn: <https://www.linkedin.com/in/prof-dr-jaspreet-singh-hira-7613291b/>

Area of Interest: Hybrid Metal Matrix Composite, Manufacturing Technology, Industrial Engineering, Fuel Technology, IC Engines.

Subject of Interest: Material Science, Manufacturing Processes, IC Engine, Thermodynamics, Automobile Engineering.

Technical Skills: XRD, SEM, Matlab, GREET Model, UTM, Vicker-hardness, Taguchi Technique, Anova, Emission Analyzer, Engine Optimization Techniques.

Soft Skills: Proficient English Language

Work Experiences:

Sr. No.	Designation	Institute Name	Type of Work	Date of Joining	Total Experience
					16 years and 5 Months
1.	Professor	Maharishi Markandeshwar University, Mullana Ambala	Academic and Research NAAC and NBA	19 Jan 2026 – Till Date	6 Months
2.	Professor and HOD	Uttaranchal University, Dehradun	Administrative, Academic and Research NAAC	25 July 2025 – Jan 2026	6 Months

3.	Professor and Dean Engineering	Raffles University, Neemrana Rajasthan	Administrative, Academic and Research NAAC work	Oct 2023- July 2025	1 Year and 9 Months
4.	Associate Professor	Raffles University, Neemrana Rajasthan	Academic and Research NAAC	Sep 2021 – Oct 2023	2 Years and 1 Month
5.	SRF	IIT Delhi	Research	Dec 2020- Sep 2021	9 Months
6.	Assistant Professor	Amity University Noida	Academic and Research NAAC	June 2015- Dec 2020	5 Years and 6 Months
7.	SRF	PEC University of Technology Chandigarh	Research	Aug 2011- June 2015	3 Years and 10 Months
8.	Lecturer	Rayat and Bahra University	Academic	July 2010- July 2011	1 Year
9.	Assistant Professor	Indo Global College of Engineering Abhipur Mohali	Academic	Aug 2007- Jan 2008	6 Months

Education Qualifications

Sr. No.	Degree	Institute Name	Branch	Result
1.	PHD 2017	PEC University of Technology Chandigarh	Mechanical Engineering	100
2.	M.Tech 2010	IIT Delhi	Specialization in Energy Studies	7.37 CGPA
3.	B.Tech 2007	IET Bhaddal Ropar Punjab (Punjab Technical University – Jalandhar)	Mechanical Engineering	61.41 %
4.	GATE 2008		Production and Industrial Engineering	AIR – 65

Detailed Information:

1. Doctor of Philosophy (Ph.D.) in Mechanical Engineering (Awarded: May 2017)
Punjab Engineering College, University of Technology, Chandigarh

Ph.D. Thesis Title: *Fabrication and Processing of hybrid Mg/(SiCp+Al₂O₃p+Grp) metal matrix composite by Gas Injection Process*

Supervisor: Prof. (Dr.) Alakesh Manna, Head of the Mechanical Engineering Department, PEC, University of Technology, Chandigarh, Email Id: kgpmanna@rediffmail.co.in, Mobile Number: 9781986570

Co-Supervisor: Dr. S.K.Mangal, Head of the Mechanical Engineering Department, PEC, University of Technology, Chandigarh, Email Id: kgpmanna@rediffmail.co.in, Mobile Number: 9876613657

Courses Completed: Mechanical Behavior of Materials, Advanced Manufacturing Process, Design of Experiments and Research Methodology

2. Master of Technology in Department of Energy Science and Engineering (Awarded: Aug 2010) Indian Institute of Technology Delhi

M.Thesis Title: *Control of NO_x emission using cooled EGR by using Karanja Biodiesel fuel in diesel engine.*

Supervisor: Prof.(Dr.) K.A.Subramanian, Head of the Department of Energy Science and Engineering, IIT Delhi

Courses Completed: Fuel Technology, Power Plant Engineering, Non-Conventional, Source of Energy, Heat Transfer, Design of IC Engine Components and Subsystems, Solar Refrigeration and Air Conditioning, Alternative fuels for Transportation, Energy, Ecology and Environment, Energy Conservation, Energy Auditing, Direct Energy Conversion, Economic and Planning of Energy Systems, Cogeneration and Energy Efficiency, Analysis of I.C.Engine Processes.

Result: CGPA obtained 7.37

3. Bachelor of Technology in Mechanical Engineering Department (Awarded: Aug 2008) Institute of Engineering and Technology, Bhaddal Ropar, Punjab; Institute affiliated to Punjab Technical University Jalandhar

Result: 61.41 %

4. GATE Qualified: AIR 65 Production and Industrial Engineering (GATE - 2008)

Patents (Title and Patent Number)

1. A Mosquito Bat Assembly/ RU/ 202211004809
2. An onboard Food Warmer Assembly/RU/202211051624
3. An Integrated Stir Casting Machine/AU/202011036189
4. An Assistive Mechanism for Adaptive Riding Positions/AU/202011003038
5. Abdomen and Oblique Twister Machine/AU/201811016525A
6. AI Based Pharmaceutical Dispensing Systems/ 443798001

Research Articles in International Journals

1. **Hira, Jaspreet**, S. K. Mangal, and Alakesh Manna. "Fabrication of hybrid Mg/(Al₂O_{3p}+ SiC_p+ Gr_p) metal matrix composite on developed gas injection liquid stir casting setup." *Arabian Journal for Science and Engineering* 40.9 (2015): 2729-2738. **IF: 3.1. (Total Citation: 11)**
2. Kamboj, K., Singh, G., Sharma, R., Panchal, D., & **Hira, Jaspreet**. (2017). Heat transfer augmentation in double pipe heat exchanger using mechanical turbulators. *Heat and Mass Transfer*, 53(2), 553-567. **IF: 2.5 (Total Citation: 39)**
3. Sharma, Rohit, Vikas Kumar, **Jaspreet Hira**, and Ravinder Kumar. "Physiological and psycho-physiological approaches for maximal carrying load estimation." *Engineering and Applied Science Research* 48, no. 5 (2021): 537-546. **Scopus Indexed.**
4. Kumar, Vikas, Kindu Oumer, Eshetu A. Merso, Rohit Sharma, **Jaspreet Hira**, and Barun Halder. "Ergonomic and Anthropometric Evaluation of Locally Manufactured Vehicle Seats." *Indian Journal of Occupational and Environmental Medicine* 27, no. 4 (2023): 338-346. **IF: 0.8**
5. **Hira, Jaspreet**, Basant Singh Sikarwar, Rohit Sharma, Vikas Kumar, and Prakhar Sharma. "Design and development of surge tank for NOx emission reduction in B20 biofuelled diesel engine." *International Journal of Vehicle Structures & Systems* 12, no. 5 (2020): 494-497. **Scopus Indexed. (Total Citation: 02)**
6. Shubbender, **Jaspreet Hira** and Parveen Berwal, "Economic and Social Impact of Traffic Congestion in Gurugram: A Comprehensive Analysis and Proposed Solutions" *International Journal of Applied Engineering & Technology*, Vol 5, June 2023, 177-191. **Scopus Indexed.**
7. **Hira, Jaspreet**, Nalin Chaudhary, Bharat Ankur Dogra, Rohit Sharma, Vikas Kumar, and Kavita Rani. "Comparison of the emissions of different biofuels with fossil fuel through the utilization of GREET Model." In *IOP Conference Series: Materials Science and Engineering*, vol. 1228, no. 1, p. 012023. IOP Publishing, 2022.

International Conference Publications

1. **Hira, Jaspreet**, Alakesh Manna, Binit Kumar Jha, Rohit Sharma, Anup Malik, and Deepak Bhutani. "The effect on the tool flank wear width V_{Bmax} during turning of Hybrid Mg/(8wt% SiC_(p)+ 2wt% Al₂O_{3(p)}+ 1wt% Gr_(p)) MMC using PVD coated cermet tool." In *IOP Conference Series: Materials Science and Engineering*, vol. 1225, no. 1, p. 012006. IOP Publishing, 2022.
2. Sharma, Aashray, Preeti Joshi, and **Jaspreet Hira**. "Utilization of nanofluids as coolant—a review." *Advances in Fluid and Thermal Engineering: Select Proceedings of FLAME 2020* (2021): 731-740.
3. **Hira, Jaspreet**, Alakesh Manna, Pushpinder Gera, Rohit Sharma, and Vikas Kumar. "Effect of machining parameters on average surface roughness Ra while turning hybrid Mg-MMC-An experimental approach." In *Journal of Physics: Conference Series*, vol. 1854, no. 1, p. 012044. IOP Publishing, 2021.
4. Bhardwaj, Shashank, **Jaspreet Hira**, Rohit Sharma, Srijit Krishnan, and Puneet K. Jain. "Adaptability in Riding Positions: Need for India." In *Advances in Interdisciplinary Engineering: Select Proceedings of FLAME 2020*, pp. 77-86. Singapore: Springer Singapore, 2021.
5. K. A. Subramanian* Nidhi, **Jaspreet Hira**, Comparative analysis study on the performance and emission characteristics of GE15M5 fuel blend and E0 on a single cylinder gasoline engine using port fuel injection systems, 12th International Symposium on Fuels and Lubricants, Emerging Trends in Fuels and Lubricants.

6. **Hira, Jaspreet**, Alakesh Manna, S. K. Mangal, and Pushpinder Kumar. "The Effect of Friction Stir Welding on the Mechanical Properties of Hybrid Mg/(SiC_p+ Al₂O_{3p}+ Gr_p)-MMC." In *Advances in Systems Engineering: Select Proceedings of NSC 2019*, pp. 737-746. Singapore: Springer Singapore, 2021.
7. Khangarot, Harshita, Vikas Kumar, Rohit Sharma, Ambrish Maurya, and **Jaspreet Hira**. "Modal Analysis of Vibratory Finishing Machine Using ANSYS Software." In *Advances in Systems Engineering: Select Proceedings of NSC 2019*, pp. 447-456. Singapore: Springer Singapore, 2021.
8. Sharma, Prakhar., **Jaspreet Hira**, and Prashant. Anand. "Development of a Surge Tank Set-up and Its Utilization in the Diesel Engine for NOX Emission Reduction." In *Emerging Trends in Mechanical Engineering: Select Proceedings of ICETMIE 2019*, pp. 187-197. Singapore: Springer Singapore, 2020.
9. **Hira, Jaspreet**, Alakesh Manna, Pushpinder Gera, Rohit Sharma, and Vikas Kumar. "Effect of machining parameters on average surface roughness Ra while turning hybrid Mg-MMC-An experimental approach." In *Journal of Physics: Conference Series*, vol. 1854, no. 1, p. 012044. IOP Publishing, 2021.
10. **Hira, Jaspreet**, Rohit Sharma, Kushal Kamboj, Vikas Kumar, and Prakhar Sharma. "NOx emission reduction in diesel engines through developed cooled EGR setup." In *Advances in Interdisciplinary Engineering: Select Proceedings of FLAME 2018*, pp. 1-13. Singapore: Springer Singapore, 2019.
11. Bhardwaj, Mohit, Shivam Mittal, Vikas Kumar, Rohit Sharma, and **Jaspreet Hira**. "Comparative model analysis of brake rotors." In *Advances in Engineering Design: Select Proceedings of FLAME 2018*, pp. 441-449. Singapore: Springer Singapore, 2019.
12. Rohit Singla, Saurabh Kumar Yadav, **Jaspreet Hira**, Augmented Reality-Based Simulation of Spring–Mass System, *Advances in Engineering Design: Select Proceedings of FLAME 2018*.
13. Alakesh Manna S K Mangal **Jaspreet Hira**, A Study during Fabrication of Hybrid Mg/(Al₂O_{3p}+SiC_p+Gr_p) Composite utilized Fabricated Gas Injection Magnesium Melting Set-up, 23rd International Conference on Processing and Fabrication of Advanced Materials.

National Conferences

1. Jaspreet Hira, Alakesh Manna and S.K.Mangal, An Investigation on Fabrication of magnesium-based metal matrix composite, *Proceedings of National Conference on Advancement and Futuristic Trends in Mechanical Engineering*, PEC University of Technology. 2014
2. Jaspreet Hira, Kushal Kamboj, Rohit Sharma, An experimental approach to indentify optimum pitch of a mechanical turbulator in DPHE, *Recent Advances in Mechanical Engineering*, PIET. 2017

Research Paper Communicated 2026

Journal Articles 2026

1. Dr.Mohit Sharma*, Dr.Jaspreet Hirab, Dr.Ashutosh Dhamijac, Dr.Anukriti Sharmad, Dr. Nidhie, Sustainable Utilization of Paddy Straw through Producer Gas Generation and Dual-Fuel Operation in a Variable Compression Ratio CI Engine (Submitted in *Biofuels Journal*).
2. Dr. Jaspreet Hira, Dr. Mohit Sharma, Dr. Anil Dhanola, Manufacturing Challenges and Solutions for Hybrid Metal Matrix Composites: A Critical Review. (About to Submitted in *Journal of Composite Materials*).

Conferences Articles Accepted 2026

1. **Hira, Jaspreet**, Mohit Sharma, Shaman Gupta, Sameer Sharma, Ashutosh Dhamija, and Anukriti Sharma. "Investigation of the mechanical properties of Mg-SiCp+Al₂O₃p+Grp Hybrid Metal Matrix composite through Argon Gas Injection Stir Casting Process- A Modern Approach." In Proceedings of the Biennial International Conference on Future Learning Aspects of Mechanical Engineering (FLAME). Amity University Uttar Pradesh, Noida, India. (Accepted).
2. **Hira, Jaspreet**, Ashwani Verma, Mohit Sharma, and Ashutosh Dhamija. "Optimization of Turning Parameters for Improved Surface Finish in Hybrid Mg/SiC/Al₂O₃/Gr MMC Using Taguchi Methodology with WC Inserts." In Proceedings of the Biennial International Conference on Future Learning Aspects of Mechanical Engineering (FLAME). Amity University Uttar Pradesh, Noida, India. (Accepted).
3. Antil, Surender, Mohit Sharma, **Jaspreet Hira**, Ashutosh Dhamija, and Anukriti Sharma. "Experimental Study on the Pilot Scale Toroidal Fluidized Bed Gasification System Utilizing Rice Straw Pellets for Producer Gas Generation." In Proceedings of the Biennial International Conference on Future Learning Aspects of Mechanical Engineering (FLAME). Amity University Uttar Pradesh, Noida, India. (Accepted).

Supervision and Research Guidance

1. Ramkesh Meena (Roll No. 2201ET009), Raffles University – 2024, Thesis Title: Method to Enhance the Charge Density of Lithium-ion Batteries and Reduction Techniques.
2. Shubhender (Roll No. 2201ET006), Raffles University – 2024, Thesis Title: Analysis of Traffic Congestion at Gurugram: Its Impact and Solutions

M.Tech. Dissertation Supervision

1. Shashank Bhardwaj (Roll No. A2326218007), Amity University – 2020
Dissertation Title: Adaptability in Riding Positions: Need of India
2. Aashray Sharma (Roll No. A2325314003), Amity University – 2020
Dissertation Title: Utilization of Nanofluids in Motor Vehicle Radiator System to Evaluate Optimum Heat Transfer Coefficient
3. Pankaj Kumar (Roll No. 22RU33302), Raffles University – 2024
Dissertation Title: Analysis of the Mechanical Properties and Testing Procedures for a Hybrid Composite Materials.
4. Manish Kumar Sharma (Roll No. 22RU33301), Raffles University – 2024
Dissertation Title: Reciprocating and Evaluation of a Wind Harvest
5. Prakhar Sharma (Roll No. A2326217015), Amity University – 2019
Dissertation Title: Reduction of NO_x Emission from a Diesel Engine Using Cold EGR

Invited Speaker

1. National Conference, Department of Mechanical Engineering, Panipat 12th–13th , December 2019

International Conference Organized:

1. FIST (Future Integration in Science and Technology) Role:
Organizing Secretary for International Conferences, Date: 08/2020

Reviewer (Journal)

1. Journal of Process Mechanical Engineering, Paper Title: "Water Absorption Behavior of Glass Fibre Reinforced Polymer Composite with Clamshell and Cenosphere Filler" Discover Applied Sciences (Springer Nature), Awarded a Reviewer Certificate for 2025 for reviewing 1 manuscript.

Member of Professional Bodies

1. Organizing Secretary, International Conference FIST (Future Integration in Science and Technology), August 2020

Professional Workshops, FDP's and Training Programs Attended

1. Faculty Development Programs (FDPs), Workshops & Training Attended
2025, DST (Rajasthan) Sponsored Five-Day National Workshop
Title: Metal Matrix Composites: Exploring New Frontiers in Manufacturing and Automation
Organizer: Anand International College of Engineering, Jaipur
Dates: March 3–7, 2025
2. One-Week Faculty Development Program (FDP)
Title: Mathematical and Computational Aspects of Industry 4.0 (MCAI 2K24)
Organizer: Haldia Institute of Technology & O.P. Jindal University
Dates: June 3–7, 2024
3. Two-Week Faculty Entrepreneurship Development Program (FEDP)
Organizer: ACIC RISE Association, Chandigarh Engineering College, Mohali
Dates: February 26 – March 9, 2024
4. Lecture Series
Title: MATLAB for Programmers
Organizer: Vigyan Gurjari in association with Parul University
Dates: July 15 – August 9, 2024
5. Three-Day Workshop
Title: Scientific Computing using MATHEMATICA
Organizer: Geethanjali College of Engineering and Technology, Hyderabad
Dates: May 23–25, 2024
2023

6. One-Week Faculty Development Program (FDP)
Title: Recent Trends of Teaching–Learning Pedagogy in Higher Education
Organizer: School of Law, Raffles University
Dates: August 13–19, 2023
2015
7. Faculty Development Program
Title: Optimization Techniques in Engineering
Organizer: Department of Mechanical Engineering, Amity University, Noida (U.P.)
Dates: June 29–30, 2015
Sponsor: Amity University
8. Faculty Development Program
Title: Faculty Development Programme on Amity
Organizer: Department of Mechanical Engineering, Amity University, Noida (U.P.)
Dates: June 12–14, 2015
Sponsor: Amity University
9. Faculty Development Program
Title: Academic Structure and System
Organizer: Amity University
Dates: August 2015
Sponsor: Amity University
2014
10. Short-Term Course
Title: Advanced and Micromanufacturing
Organizer: Department of Mechanical Engineering, PEC University of Technology, Chandigarh
Dates: June 2–6, 2014
Sponsor: TEQIP–II (MHRD)
2013
11. Short-Term Course
Title: Advanced and Micromanufacturing
Organizer: Department of Mechanical Engineering, PEC University of Technology, Chandigarh
Dates: June 3–7, 2013
Sponsor: TEQIP–II (MHRD)

Dr. Jaspreet Hira
Chandigarh
Date: 01.08.2026